Verify Everything, Learn From Every Round

Measured, equivalence-gated RTL PPA optimization

sha512: ADP 0.787, timing –97 ps → +235 ps (MET), LEC-proven —
and the agent found the same trick on its own

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The problem, and what actually scores

Given: 7 IPs (async_fifo, sha512, NVDLA, OpenTitan aes/ascon/kmac/prim), Verilator/iverilog testbenches, a Yosys+OpenSTA ASAP7 flow. **Rewrite RTL to improve PPA.**

Evaluation (per the organizers): functional correctness with the existing testbenches first; then PPA after Yosys, plus **LLM calls and token cost**.

Why this is hard (Alpha-RTL, ICCAD'24 lineage): on async_fifo, **every published LLM-rewriting method scored zero** — no compilable, correct rewrite. Unverified "optimizations" are noise; the synthesizer absorbs or breaks most local edits.

Our headline metric: baseline-normalized **Area-Delay-Product ratio**, functional- and equivalence-gated, token-conscious.

Thesis: nothing counts until it survives five gates and a measurement

```
propose (k strategies)          ← prompt ladder + learned playbook + STA feedback
   │
verify: lint → compile → TB gate → yosys LEC (+async2sync) → dual-instance
   │                               differential simulation
measure: Docker Yosys synth + OpenSTA (parallel APFS-clone workspaces)
   │
select: Pareto frontier / ADP, accept only measured strict improvement
   │
learn: Thompson parent sampling · ACE playbook · reflector
```

Accepted ⇒ formally equivalent AND measurably better. Everything else is rejected with a typed reason that feeds the next round.

Headline result: sha512, LEC-proven

Metric	Baseline	Optimized (exp2)	Δ
WNS	**−97.30 ps** (violated)	**+235.36 ps (MET)**	+332.66 ps
Area	3984.2 µm²	3960.3 µm²	−0.6%
Cells	29,307	28,955	−1.2%
Power (our flow)	2.92 mW	2.23 mW	−24%
ADP ratio	1.000	**0.787**	**−21.3%**

Technique: serial N-operand addition chains → balanced adder trees (mod-2^w associativity ⇒ provably equivalent). **yosys LEC: PROVEN. Differential sim: PASS.**

Plus: a gate-passing, LEC-proven async_fifo variant — on the IP where published methods score 0.

Honest measurement as a feature: the aes baseline story

While onboarding OpenTitan aes we found the shipped flow **silently broken**:

- `generated/all_modules.v` duplicates all 84 per-module files
- under `SKIP_SV2V=1`, yosys hits re-definition ERRORs — **masked by the run script's success banner**; no netlist is ever written
- the recorded aes baseline (–1083.58 ps) is **invalid**; clean baseline = **–848.15 ps**

Our workspaces auto-drop the duplicate file (contest repo untouched); every baseline we report is re-derived from a pristine clone through one flow. **Toolkit bug #1 of 3 we found** (the other two in the NXP toolkit — see our NXP deck).

Live with Gemini (Jul 12): the empty-response mystery

First real rounds: proposals came back **empty** while consuming ~20k tokens each.

Probe on a real prompt:

```
finish_reason: MAX_TOKENS   parts: 0
thoughts_token_count: 62,913   candidates: 2,619
```

`gemini-3-flash-preview` **thinks by default** and burned the entire cap on thoughts.

Fixes now in the agent: explicit `max_output_tokens` + bounded `thinking_config`; token accounting uses `total_token_count` (thoughts are billed!); empty-text retry; raw responses archived for postmortems; fence-parsing fallback (models ignore format instructions ~2/3 of the time).

Live: the agent's first verified accepts

```
[sha512] round 1: parent=baseline, tag=arith-carry-chain, k=3
  reject 4f97bf841af0 [balanced-tree]  duplicate
  reject 78633e719135 [carry-save]     gate-fail
  ACCEPT 3939ed94c02f [arith-arch]     improves perf, regresses none | ADP 0.898
```

then, two rounds later, **composition** via Thompson parent sampling:

```
[sha512] round 2: parent=3939ed94c02f (arith-arch)
  ACCEPT e7fee825c983 [restructure-select]  improves perf,area | ADP vs parent=0.985
```

→ model-generated line at **≈0.884 vs baseline**, every layer verified.

Banked exp2 (**0.787**) remains global best — and is what the agent emits.

The duplicate moment (validation money-shot)

In the same round, Gemini proposed the balanced-adder-tree rewrite — **independently reinventing our hand-derived winning technique.**

```
reject 4f97bf841af0 [balanced-tree] duplicate: duplicate
```

- the **fingerprint dedup** recognized it instantly — **zero wasted synthesis, zero wasted verification**
- the strategy playbook and the model converge on the same transformations — evidence the approach generalizes rather than being hand-crafted luck

The learning loop is real

Playbook (ACE-style): 23 measured rules. From measured no-ops (exp3/exp4):

> *AVOID: local restructuring (mux folding, operand swap) — ABC absorbs it; only global algebraic changes survive synthesis.*

Live-learned (Jul 12), written by the reflector after a real gate-fail:

> *(sha512) Manual carry-save (3:2 compressor) insertion can lead to gate-level synthesis flow failures.*

Rules are voted helpful/harmful per round (Beta posterior) and injected into future prompts — failures compound into guidance, not just logs.

Hidden testcases: drilled, not hoped for

Auto-discovery builds an IPSpec from repo conventions (env.sh, filelists, SDC, TB runners): fixtures match the hand registry 3/3; aes/kmac/prim/**NVDLA (323 sources)** onboard zero-config.

Cold-start drill (`test_cold_start.py`, repeatable, 6/6): plant an unseen IP → discover → fresh baseline → propose → verify → measure → decide.

The drill **caught a real scoring hazard**: discovery keyed IPs by `env.sh DESIGN_NAME`, so a hidden testcase colliding with a known name would have been scored against the **wrong cached baseline**. Fixed (location-derived keys) — found by drilling, not luck.

Submission artifact: `--emit-best`

Every campaign ends with a drop-in artifact — even a no-improvement one (explicit manifest beats silence):

```
{
 "ip": "sha512",  "result": "optimized",
 "changed_files": ["sha512/src/rtl/sha512.v", "..." ],
 "baseline_ppa": {"area": 3984.2, "setup": -97.30, ...},
 "best_ppa":     {"area": 3960.3, "setup": 235.36, ...},
 "adp_vs_baseline": 0.787,
 "verification": "full 5-layer at acceptance: lint, compile, TB gate,
                 yosys LEC (+async2sync), dual-instance differential sim",
 "llm_calls": 7, "llm_tokens_approx": 202931
}
```

Files land in repo-relative layout — the evaluator (or a human) drops them straight in.

Token economics

Mechanism	Effect
Budget regimes (70/30/10) + `<budget>` line in prompts	model knows its remaining budget
Plateau stop (frontier ADP flat)	no burn on converged IPs
Proxy pre-filter (yosys stat/ltp) + exact-fingerprint dedup	cheap rejection before synthesis
Measured round cost	~112k tokens (k=3, thinking included), ~2.3 min
Thinking budget capped	unbounded = 63k thought tokens for zero output

All calls/tokens ledgered per round; usage extractable exactly as the organizers evaluate.

Model-agnostic interface

- `--model vertex` — google-genai; auto-detects **Vertex Express Mode** key (the organizers' eval path, AgentSetup.md) or AI Studio key; retry ladder on 429/5xx
- `--model endpoint` — NXP-runner-style HTTP model service, in case eval fronts Vertex
- `--model stub` — full offline loop for CI (replayable variants)
- `--model-name`, `--temperature/--top-p` — the decoding-config sweep + model-mixing knobs

Mock-SDK test exercises the real code path with no network: **13/13**
(both key modes, usage accounting, retry behavior, endpoint protocol).

By DAC (agents updatable through Jul 26)

- **Decoding-config sweep** (temp 0–0.4 / top_p 0.4–0.7) + **model mixing** (pro for proposals, flash for repairs) — on the Jul-19 unlimited-Gemini accounts
- **aes GF/tower-field + kmac Keccak-θ campaigns** — the ABC-resistant headroom our survey identified
- NVDLA baseline; GEPA offline prompt evolution
- Live-testcase runbook: triage tree, budgets, panic modes (in repo)

Summary — every claim, one command

Claim	Reproduce with
Loop offline e2e	`python3 -m ppa.controller --ip async_fifo --rounds 1 --k 1 --model stub`
Fresh baseline (Docker)	`python3 -m ppa.evaluate --ip <name|path> --baseline`
5-layer verify	`python3 -m ppa.verify --ip sha512 --variant-dir ../exp2_sha512_balanced`
Hidden-testcase drill	`python3 test_cold_start.py` → 6/6
Model interface	`python3 test_model_iface.py` → 13/13
Real campaign	`python3 -m ppa.controller --ip sha512 --rounds 3 --k 3 --model vertex --emit-best` `…`

Repo: github.com/chelsea85/chip-convergence-iclad26 (fresh-clone verified)

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